Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/16/2013
PI Name:	Kearney, Anne Ph.D.		
Project Title:	Environmental Factors Evidence Review		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98101	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2012	**End Date:**	04/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	Spaceflight environments pose significant psychological risks and maximizing and maintaining mental functioning and psychological wellbeing during an extended stay in these confined spaces, with sensory monotony, social isolation, constant threat of danger, and extreme physical demands is a challenge. This task provides: (1) a literature review of the psychological and behavioral risk factors in long duration spaceflight environments and explores possible countermeasures for risk mitigation; and, (2) an operational assessment of needs for future long duration and distance exploration missions derived from current practices as perceived by Subject Matter Experts (SMEs). Interview Objectives The literature review completed as the first part of this project current literature has found that much can be learned		

Task Description:	about psychological and behavioral risk factors and countermeasures in spaceflight, analogue ICE environments, and from the broader environmental psychology literature. However, there is relatively little hard data on the application and effectiveness of countermeasures in spaceflight environments, particularly with respect to behavioral and psychological aspects of long duration and/or distance spaceflight. In addition, questions remain regarding the applicability of specific countermeasure technology in the spaceflight environment. Because of these unanswered questions, it is important to go beyond the existing literature and gather input and insight directly from subject matter experts through a series of interviews. The specific objectives of the interviews are to gather expert insight on: (1) environmental risk factors for the behavioral and psychological risks specific to long duration and distance spaceflight; (2) current countermeasure technology and design (e.g., related to environmental modifications such as visual screens, virtual reality, onsite plant production, stowage systems to reduce visual clutter, in-flight support mechanisms in the context of communication delays, minimum requirements for and constraints on private space); (3) issues related to countermeasure implementation (e.g., design constraints), including the feasibility of technologies being space ready in relation to the NASA Roadmap. Ultimately, these efforts will identify specific research requirements in environmental risk factors and countermeasures to solicit a well scoped research topic that will reduce risk. Interview Methodology Interviews will be conducted by phone with approximately 10 subject matter experts. Interviews will last no more than one hour and will be semi-structured, consisting of some general questions (asked of all interviewees) along with expert-specific questions. Prior to the interview, experts will receive a brief email providing a short overview of the project and a description of the purpose of the interview. In addition, experts will be invited to send the interviewer any documentation or other materials that might help inform the interview and ensure efficient use of the interview time. Subject Matter Experts (SMEs). The following experts would be interviewed: spaceflight doctor, Antarctic station designer, space human factors designer, human factors engineer, space architect, avionics system architect, and a space exploration vehicle engineer, as well as a BHP Operations SME with expertise in psychological support and communication. Interview Questions. All experts will be asked for their opinions on behavioral, psychological, and environmental risk factors in long duration spaceflight and about effective environmental countermeasures for mitigating behavioral health and performance risks. Following these general questions, each interview will be tailored to the specific SME, focusing on the themes outlined above. Data Analysis. A content analysis will be performed on data related to common questions and themes will be identified. Responses to SME specific questions will be synthesized and integrated into a final report.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	
Task Progress:	Decision Point is in work regarding this task. Results have informed the Net Habitable Volume effort between the SHFH and BHP. [Ed. note 9/16/13: information from PI's Task Book Initial Entry Form, provided September 2013]
Bibliography Type:	Description: (Last Updated: 03/03/2016)
Abstracts for Journals and Proceedings	Kearney A. "Countermeasures for Behavioral and Psychological Risk Factors during Long Duration Spaceflight." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013